An Optimization of IMU Sensors-based Approach for Malaysian Sign Language Recognition

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***Abstract*—Malaysian Sign Language (MSL) has been the main communication medium for Malaysian deaf people but only a small portion of ordinary people can comprehend this MSL. A common approach to convey the meaning of sign language is by using sign language recognition (SLR) method. A video-based SLR approach has been researched extensively but still heavily vulnerable to environment error and suffer from portability issue. Conversely, a sensor-based SLR solution can mitigate these problems at the cost of having bulky design due to the number and placement of the sensors required to achieve sufficient accuracy. Therefore, in this paper, we investigated the optimal number and placement of the inertial measurement unit (IMU) sensors required to perform an accurate conversion. Experimental results demonstrated our proposed optimization process is capable of achieving more than 98% of recognition accuracy. This could be a benchmark for a less bulky and leaner design of sensor-based hand glove as well as overcoming the usability and portability issues.**

Keywords—Malaysian Sign Language, Sign Language Recognition, IMU sensor optimization.

I. Introduction

Until January 2020, there are more than twenty-four thousand deaf people registered with the Society Welfare Department of Malaysia [1]. For these people, Malaysian Sign Language (MSL) has become the most important medium to connect themselves with society. However, sign language is not yet common practice to the majority of normal people, thus, it is difficult to be comprehended. Therefore, the sign language recognition (SLR) technique has been researched extensively to help this group of deaf people [2-10].

SLR can be classified into two approaches; video-based [2],[3] and sensor-based solutions [4-7]. While having some advantages, the video-based solution has quite a critical drawback where it is extremely sensitive to the background noise and lighting condition. Moreover, the necessity of camera deployment has severed the portability issue. On the other hand, a wearable sensor-based solution can overcome these issues well. Many solutions have been proposed previously by using various types of sensors such as ElectroMyoGraph (EMG) [4], flex sensors [5] and inertial measurement unit (IMU) sensors [6]. However, despite its advantages, the sensor-based solution also suffers from the usability issue whereas the user needs to wear a hand glove before they can communicate with others. Since the prediction accuracy is more or less depending on the number of sensors deployed at the hand glove, the design cannot be avoided from being bulky and limit user mobility [6].

In this paper, we research on the optimization part of IMU sensors in order to obtain sufficient accuracy while reducing the number of sensors deployed on the glove. The significance of our research can assist the industry in developing more elegant and simple hand glove design such as converting into a ring-type in order to improve the usability problem.

II. Previous Works

A. Malaysian Sign Language

MSL or Bahasa Isyarat Malaysia (BIM) has been widely used by Malaysian deaf people for daily interaction. Even though MSL might have been localized in some region or community, the standard MSL is still understandable by most of the deaf people. Therefore, our priority in this current study is only focused on the standard MSL defined by the Malaysian Federation of the Deaf (MFD), a qualified organization that manages and maintains MSL of Malaysia.

Previous work on the MSL has been done by using flex sensors placed at each finger and IMU sensor at hand [5]. However, MSL has many gestures that do not merely rely on finger flexion, for example, “window”, “house”, “door” or “what” as depicted in Fig. 1. Therefore, the flex sensors are unable to compromise with these kinds of movements due to no stretching at the fingers to provide changes for the sensor’s difference voltage. Moreover, depending only on one IMU sensor at hand is inefficient for a large dataset.

Not all the fingers play an equivalent role while performing the gesture in the sense that some of the fingers may be less important to be acknowledged for some specific gestures. For example, the gestures in Fig. 1 can be successfully distinguished by using sensors on the thumb and pointer finger. This condition can be different if we look into other sets of gestures. Hence, any SLR system is compulsory to be verified with a large dataset. In addition, for MSL, right-hand fingers are mainly used to perform the gesture. However, there are gestures such as “sit down” and “chair” which have similar movements at the right-hand side but different at the left-hand side. Therefore, a combination of both left and right-hand sides are necessary to support the large MSL datasets effectively.

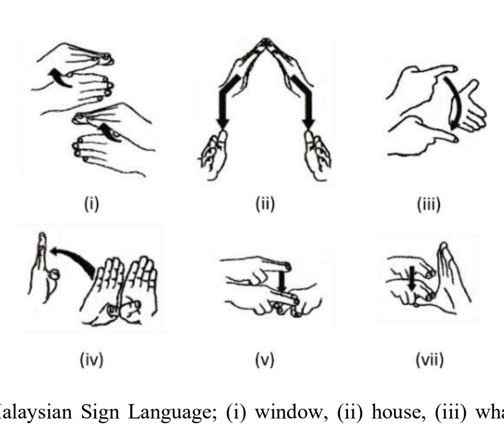

Fig. 1. Malaysian Sign Language; (i) window, (ii) house, (iii) what, (iv) door, (v) sit down and (vi) chair.

B. IMU Sensors

An IMU sensor consists of a 3-axis accelerometer, a 3-axis gyroscope, and a 3-axis magnetometer to measure the acceleration of a movement, angular velocity, and direction, respectively. IMU sensors have been widely attached on top of any devices such as hand glove or robot for the purpose of positioning and orientation tracking due to their higher sensing accuracy, low cost, and low power consumption [6], [7].

Some of the previous works proposed the fusion of EMG and IMU sensors to improve accuracy [7]. However, wearing a large size on-the-shelf EMG sensor is less efficient in daily life where it can affect the usability and portability of the user. Thus, this work only focused on the optimization of IMU sensors to obtain sufficient accuracy.

III. DATA PROCESSING AND OPTIMIZATION

The IMU sensor-based hand glove has been designed and developed as depicted in Fig. 2. The hand glove is comprised of Raspberry Pi zero (RPIv0) as a controller and six LSM9DS1 IMU sensors located at each finger and hand. The IMU sensors are connected via four pins of 0.5mm pitch FFC cable to the controller using the I2C protocol. A TCA9548A I2C multiplexer is integrated to switch the channel between six sensors during the data acquisition process. To support the RPIv0 power up requirement, a 500mA 5V power boost module is used to step up the current and voltage from a 300mA 3.7V LiPo rechargeable battery. The device can operate approximately up to 4 hours in an active mode.

A. Data Acquisition, Pre-processing, and Classification

The time-domain data of accelerometer and gyroscope are collected using a sliding window-based approach with a sampling frequency of 952MHz. The data is stored in the temporary buffer and updated at every 5ms. We extracted the mean and root mean square (RMS) values from the buffer and feed them into the Recurrent Neural Network (RNN) classifier. The structure of our network consists of 72 numbers of inputs, 100 hidden state, and 51 outputs for each RNN stage, as depicted in Fig. 3. For the feedforward operation, we sampled the sensory data and fed it into each of the RNN stages. Then the output of each RNN stage is combined inside the fully-connected layer to produce a prediction. As for the backpropagation, we performed the training using Adagrad optimizer with a learning rate of 0.1. As a measure of generalization, we adopted the dropout, L2 loss, and early stopping technique to further increase the validation and testing accuracy.

B. Optimization Procedures

Since most of the gestures in MSL are essentially performed using the right-hand side, we initiated the optimization process using right-hand side fingers. Then, we added the left-hand side finger to improve accuracy. To avoid repetitive training activity to find the right combination, we simplify the optimization processes with the following 5 steps, where the final step is required for cross-validation purposes. We used the F1 score as a metric of comparison for the evaluation. The following shows the 5 steps taken:

1) Execute training for each IMU sensor individually: To obtain the initial accuracy value, we performed training for each sensor separately by turning on and off other sensors that are not in use. Then we recorded the result and drew confusion matrices based on the individual result.

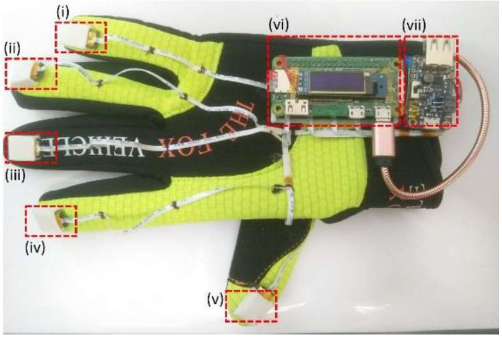

Fig. 2. IMU sensor-based hand glove; (i-v) IMU sensors, (vi) RPIv0 Controller with IMU sensor and (vii) LiPo battery and rechargeable module.

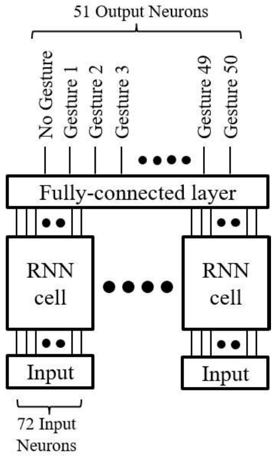

Fig. 3. RNN structure.

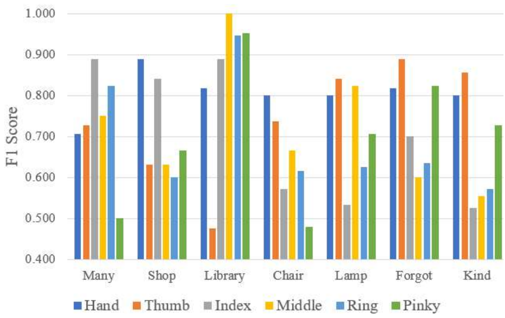

Fig. 4. F1 score for each sensor.

2) Compute F1 score and set the reference sensor: By using the confusion matrix obtained in step (1), we can compute the recall and precision as well as the F1 score for each sensor. Then we set the highest averaged F1 score sensor as a reference sensor.

3) Normalize other IMU sensors: To determine which sensors may provide a right combination with the reference sensor, firstly, we need to normalize the F1 score value of other sensors with respect to the reference sensors. Then we plot the normalization graph to visualize which sensors may have the highest normalization value. The reason is as depicted in Fig. 4, we expected that the sensors with a complementary behavior can support each other well rather than the sensors with higher individual F1 scores. In other words, if the reference sensor has a higher F1 score at any particular gesture, the other sensor does not necessarily have the same higher F1 score. On the contrary, the other sensor must have a higher F1 score when the reference sensor has a lower F1 score at any particular gesture. Therefore, from the normalization process, we can determine which sensors may have complementary behavior compared to the reference sensors. This is the key to our optimization procedure.

4) Retrain the IMU sensors combination: As we found the sensor with the highest normalization value, we performed the training by combining it with the reference sensor.

5) Repeat the training with other combinations for cross-validation: For a cross-validation purpose, we execute the training again with the second and third highest normalization value sensor and record the subsequent results.

IV. Experimental results and discussion

A. Experiment Settings

We have collected experimental data from 2 expert tutors from Society Deaf of Johor (SDJO) and 30 students from Universiti Teknologi Malaysia (UTM). Each subject performed randomly 25 sign languages out of 50 MSL that we already decided. The data collection session for each subject has been conducted in three phases; pre, action, and post. At the pre-stage, the subject has to sit down straight on the chair with both hands on their tight for 3 seconds. Then, he or she has to perform each gesture within 5 seconds during the action stage and put down the hand back on the tight upon completion. During the post-stage, the subject has to perform the same procedure as in the pre-stage to ensure the sensors reading have been initialized.

The subject has to wear our designed hand glove (as shown in Fig. 2) at both hands. We have developed the client viewer using Android and deployed at the tablet. The client viewer is used to manage the data collection sequence. Therefore, the subject has to properly follow the instructions appear at the tablet, for example, which gesture to perform and when the subject has to perform the gesture.

The collected data is imported to the database for further analysis. We developed the RNN model using the Tensorflow framework for data training. As for the data splitting, we divided the data using a 7:1:2 ratio to separate between training, validation, and testing.

B. Experiment Results

Initially, before reducing the number of sensors, we performed the training using all sensors enabled at the right-hand side to determine the confidence level of our datasets and model as depicted in Fig. 5. The training and validation result shows that our model is capable of learning and achieving high accuracy which is more than 97% for the case of all sensors enabled at the right-hand side. Then, we followed the optimization steps to perform the training for each of the sensors on the right-hand side.

It was quite difficult to decide which sensor may have a significant result on the classification for our pre-defined gestures. However, from the training, we determined that sensors at thumb and hand have relatively higher accuracy and F1 score compare to others which means we can decide either one of both sensors as our reference sensor. This is because the neural network can successfully extract some hidden pattern of the sensors readings to perform a correct prediction. Since thumb sensor consistently has higher prediction accuracy value than hand sensor, therefore, we set the sensor at thumb as our reference sensor.

Then we normalized other sensors with reference to the sensor at the thumb. Subsequently, we accumulated the normalization score to determine which sensor has the highest normalization value. As depicted in Fig. 6 for some extreme cases, we observed that the sensor at the index finger has the highest accumulated normalization value. In other words, the sensor at the index finger has a complementary behavior with respect to the sensor at the thumb. This is true if we look into some other gestures as depicted in Fig. 4, whereas for a particular gesture like "many", the sensor at index fingers shown a higher F1 score reading compare to the thumb. Whilst for "lamp", the direction went into the opposite way.

In order to invalidate our optimization method, we run training using other combinations as well and compare the results as depicted in Fig. 7. From the result, we notice that the addition of the sensor at hand to the combination of the sensor at index finger and thumb does not improve the accuracy so much. This finding proved the hypothesis of if the sensor does not complementary behaved to the current existed sensor, it will not be able to produce a significant effect on the total

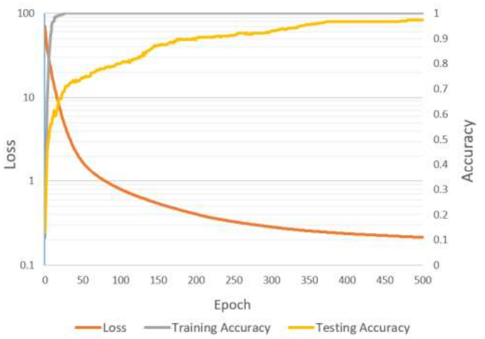

Fig. 5. Loss and accuracy during training and validation for the right-hand side all sensors enabled case.

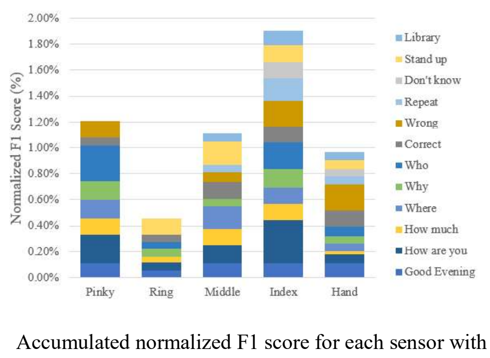

Fig. 6. Accumulated normalized F1 score for each sensor with reference to the sensor at the thumb.

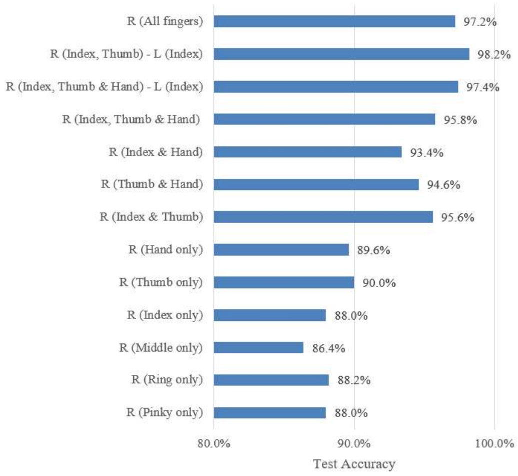

Fig. 7. Test accuracy result for each sensor and combined sensors.

accumulated accuracy even though it may have a higher F1 score individually.

To further increase the accuracy, we attempted to add the index finger on the left-hand side into the combinations. From the testing result, we can observe that the accuracy value increased significantly by adding one finger at the left-hand side. This is true for some cases such as “chair” and “sit down” which has the same right-hand gesture, but a different left-hand gesture. Overall, by combining sensors at the index finger and thumb on the right-hand side with the index finger on the left-hand side, we are able to achieve accuracy of 98.2%.

CONCLUSION

The paper intended to optimize the IMU sensor-based approach for MSL gestures to improve the usability and portability issue. From the optimization, we can observe that we are able to achieve higher accuracy by more than 98% by just placing the sensors at the specific placement which has a complementary behavior to each other. For our datasets, the suitable sensor placements are at the index finger and thumb on the right-hand side and also index finger at the left-hand side. As for other datasets, other fingers may have a significant effect, thus, they need to be validated.

From the findings, the hand glove design can be simplified and improved by reducing the number of sensors deployed while gaining sufficient accuracy for the recognition. This study can assist the industry to enhance the usability and portability issue of the sensor-based hand glove which they probably can convert the SLR system into a ring-based solution to acquire data from particular fingers that have a significant effect on the recognition process.

For future work, we will attempt to increase the size of datasets use and perform the optimization using this method. The result might be similar, or it might need some tuning on the sensor placement to achieve sufficient higher accuracy.

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REFERENCES

[1] (2020, Jan.) Formal website of the Malaysian Social Welfare Department. [Online]. Available: http://www.jkm.gov.my

[2] Suharjito, H. Gunawan, N. Thiracitta, and A. Nugroho, “Sign Language Recognition Using Modified Convolutional Neural Network Model,” 1st 2018 Indones. Assoc. Pattern Recognit. Int. Conf. Ina. 2018 - Proc., pp. 1–5, 2019.

[3] S. S. Kumar, T. Wangyal, V. Saboo, and R. Srinath, “Time Series Neural Networks for Real Time Sign Language Translation,” *Proc. - 17th IEEE Int. Conf. Mach. Learn. Appl. ICMLA 2018*, pp. 243–248, 2019.

[4] R. Fatmi, S. Rashad, and R. Integlia, “Comparing ANN, SVM, and HMM based Machine Learning Methods for American Sign Language Recognition using Wearable Motion Sensors,” *2019 IEEE 9th Annu. Comput. Commun. Work. Conf. CCWC 2019*, pp. 290–297, 2019.

[5] A. Z. Shukor, M. F. Miskon, M. H. Jamaluddin, F. Bin Ali Ibrahim, M. F. Asyraf, and M. B. Bin Bahar, “A New Data Glove Approach for Malaysian Sign Language Detection,” in *Procedia Computer Science*, 2015, vol. 76, pp. 60–67.

[6] B. Fang *et al.*, “Dynamic gesture recognition using inertial sensors-based data gloves,” *2019 4th IEEE Int. Conf. Adv. Robot. Mechatronics, ICARM 2019*, pp. 390–395, 2019.

[7] S. Shin, Y. Baek, J. Lee, Y. Eun, and S. H. Son, “Korean sign language recognition using EMG and IMU sensors based on group-dependent NN models,” *2017 IEEE Symp. Ser. Comput. Intell. SSCI 2017 - Proc.*, vol. 2018-Janua, pp. 1–7, 2018.

[8] A. Srisuphab and P. Silapachote, “Artificial neural networks for gesture classification with inertial motion sensing armbands,” IEEE Reg. 10 Annu. Int. Conf. Proceedings/TENCON, pp. 1–5, 2017.

[9] D. Naglot and M. Kulkarni, “ANN based Indian Sign Language numerals recognition using the leap motion controller,” *Proc. Int. Conf. Inven. Comput. Technol. ICICT 2016*, vol. 2, 2017.

[10] R. Siriak, I. Skarga-bandurova, and Y. Boltov, “Deep Convolutional Network with Long Short- Term Memory Layers for Dynamic Gesture Recognition,” *2019 10th IEEE Int. Conf. Intell. Data Acquis. Adv. Comput. Syst. Technol. Appl.*, vol. 1, pp. 158–162, 2019.